

Buck Converter — Power Management Assignment

ADI Internship

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Design targets: $V_{in} = 6\text{ V}$, $V_{DD} = 1\text{ V}$, $F_{sw} = 1\text{ MHz}$, $V_{out} = 1\text{ V}$, $I_{out} = 10\text{ A} \rightarrow 100\text{ A}$

Part 1 — Ideal-Switch Buck Converter

1. Component Design

1.1 Load Resistance (RL)

The load resistor sets I_{out} for a fixed $V_{out} = 1\text{ V}$:

$$RL, max = \frac{V_{out}}{I_{out}} = \frac{1V}{10A} = 0.1\ \Omega \quad (I_{out} = 10\text{ A})$$
$$RL, min = \frac{V_{out}}{I_{out}} = \frac{1V}{100A} = 0.01\ \Omega \quad (I_{out} = 100\text{ A})$$

RL range: 0.01 Ω to 0.1 Ω

1.2 Inductance (L)

Duty cycle (ideal buck):

$$D = \frac{V_{out}}{V_{in}} = \frac{1V}{6V} = \frac{1}{6} \approx 0.1667$$

Peak-to-peak inductor ripple relation, solved for L:

$$L = (V_{in} - V_{out}) \cdot \frac{D}{\Delta I_{pk} - pk \cdot F_{sw}} = \frac{(6 - 1) \left(\frac{1}{6}\right)}{1 \times 10^6} \approx 0.833\ \mu H$$

L = 0.833 μH (833.3 nH)

1.3 Capacitance (C)

$$\Delta V_{out} = 1\% \times V_{out} = 10\text{ mV}$$

$$\Delta V_{out} = \Delta I_{pk} - \frac{pk}{8 \cdot C \cdot F_{sw}} \Rightarrow C = \Delta I_{pk} - \frac{pk}{8 \cdot \Delta V_{out} \cdot F_{sw}} = 1 / (8 \times 0.01 \times 10^6) = 12.5\ \mu F$$

C = 12.5 μF

1.4 Summary

Parameter	Formula	Value
RL	V_{out} / I_{out}	$0.01\ \Omega - 0.1\ \Omega$
L	$(V_{in} - V_{out}) \cdot D / (\Delta I_{pk} - pk \cdot F_{sw})$	$0.833\ \mu H$
C	$\Delta I_{pk} - pk / (8 \cdot \Delta V_{out} \cdot F_{sw})$	$12.5\ \mu F$

2. Schematic

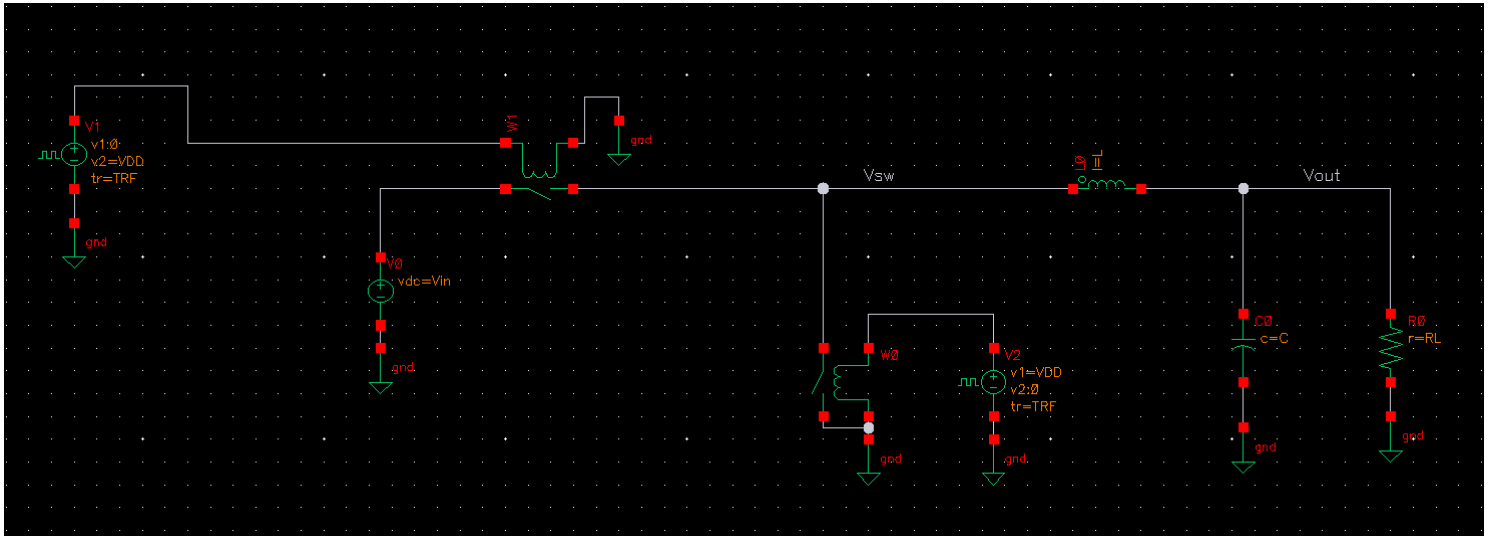


Figure 1. Buck converter built with ideal switches from analogLib. High-side and low-side switches are driven by complementary pulse sources ($V_{on} = VDD = 1\text{ V}$, $V_{off} = 0\text{ V}$, $R_{on} = 1\text{ }\mu\Omega$, $R_{off} = 1\text{ T}\Omega$), followed by the LC output filter and load R_L .

3. Vout and IL Waveforms at $I_{out} = 100\text{ A}$

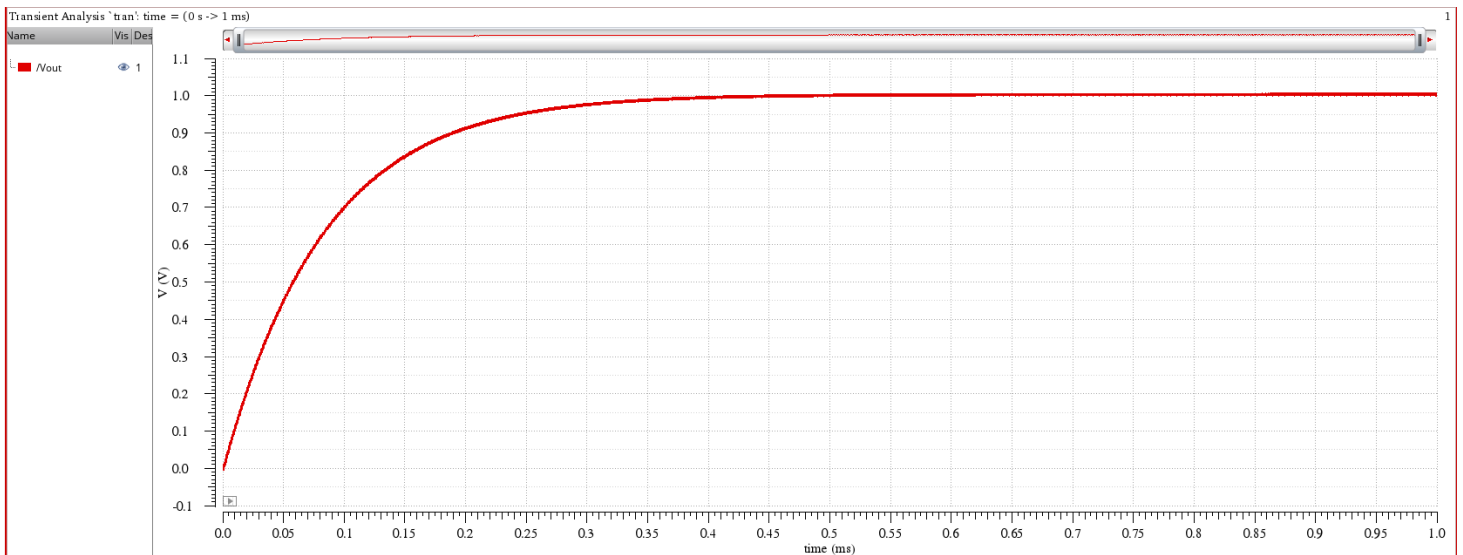


Figure 2. V_{out} transient response from start-up to steady state. X-axis: time (s). Y-axis: V_{out} (V). Condition: $I_{out} = 100\text{ A}$ ($R_L = 0.01\text{ }\Omega$), ideal switches.

Comment: V_{out} settles cleanly to the 1 V target with no steady-state error, as expected for ideal (lossless) switches.

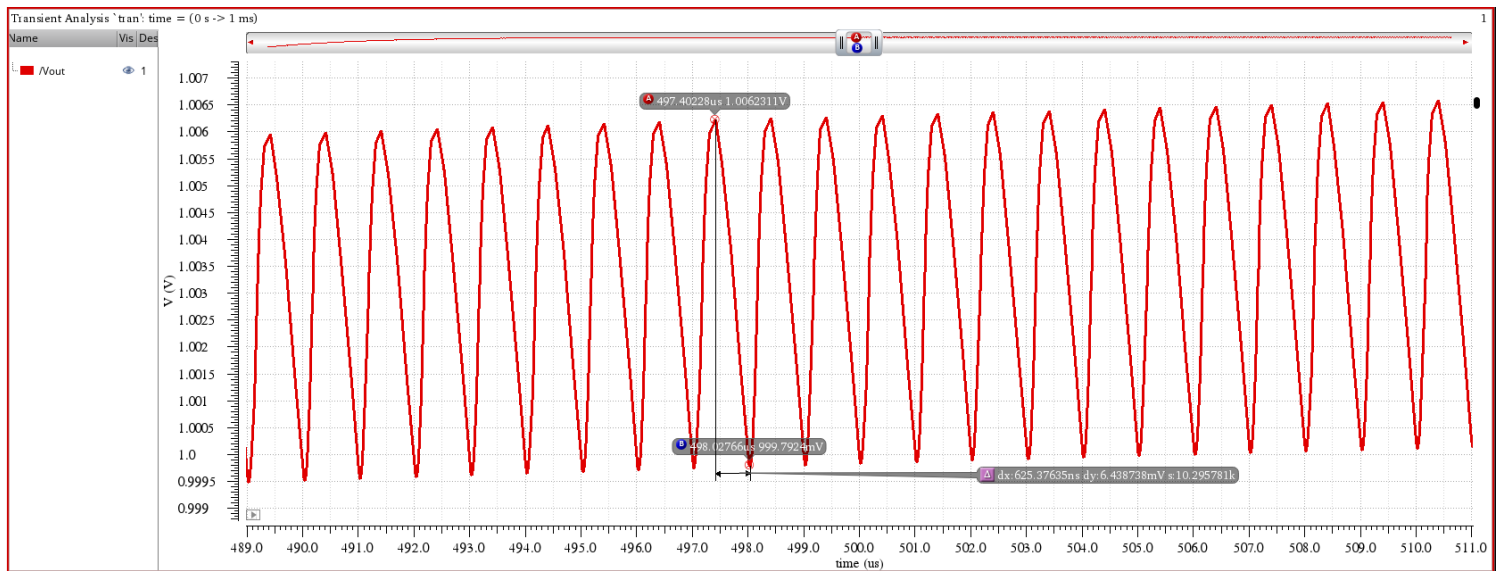


Figure 3. Zoomed steady-state V_{out} ripple over one switching period. X-axis: time (s). Y-axis: V_{out} (V). Condition: $I_{out} = 100$ A.

Comment: $\Delta V_{out,pk-pk} \approx 6 \text{ mV} < 10 \text{ mV spec (1\% of } V_{out})$. Ripple meets spec with margin because the simulated C is slightly larger than the minimum required value.

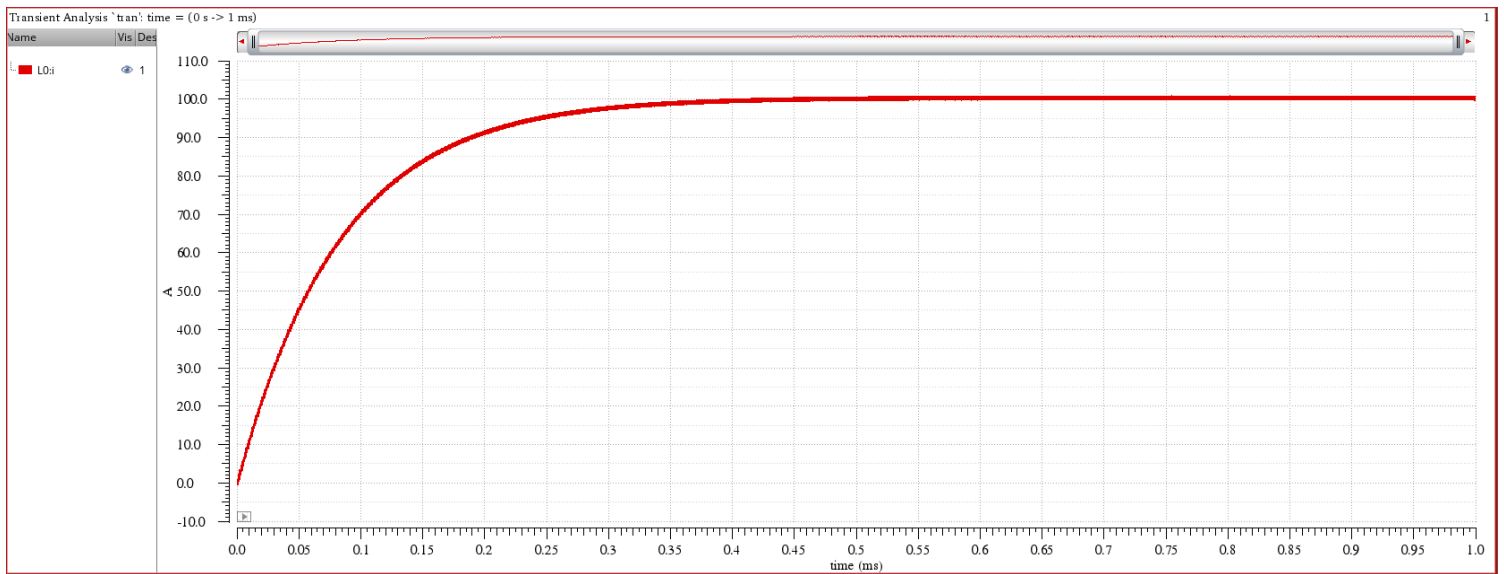


Figure 4. Inductor current I_L transient response. X-axis: time (s). Y-axis: I_L (A). Condition: $I_{out} = 100$ A, ideal switches.

Comment: I_L settles to a 100 A average, matching the load current exactly (no loss in the ideal path).

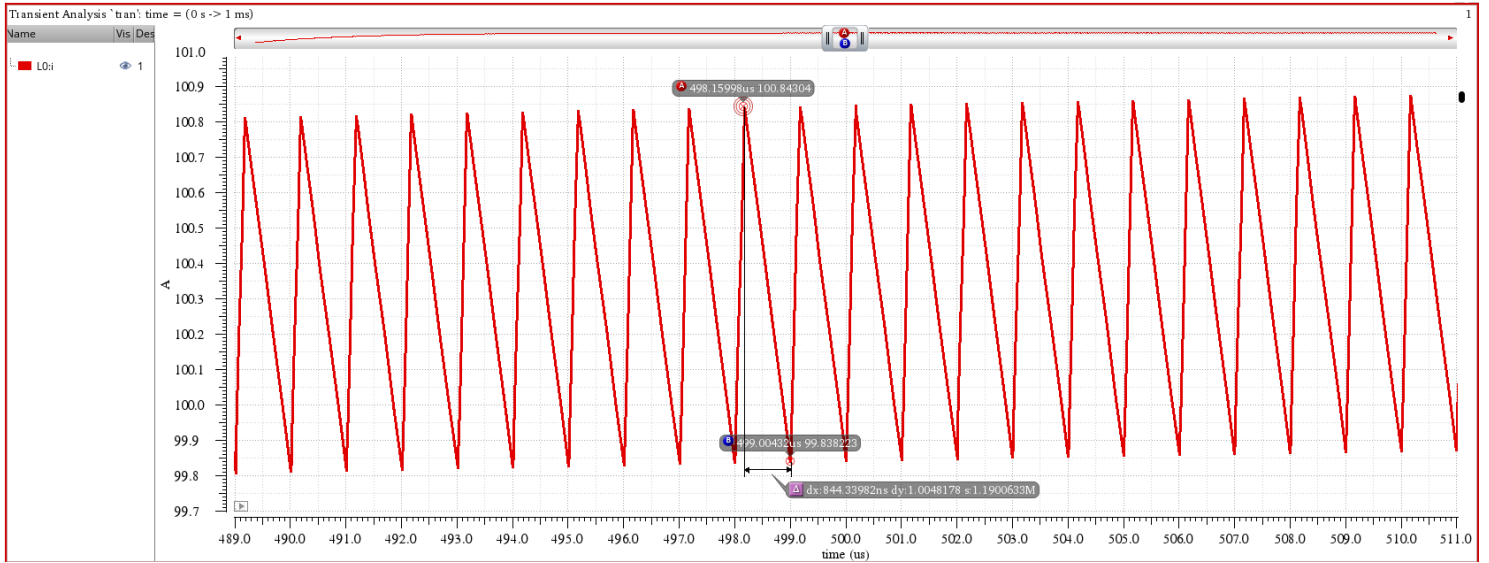


Figure 5. Zoomed steady-state IL ripple over one switching period. X-axis: time (s). Y-axis: IL (A). Condition: $I_{out} = 100$ A.

Comment: $\Delta I_{L,pk-pk} \approx 1.004$ A ≈ 1 A, matching the 1 A design target and confirming the $L = 0.833$ μ H calculation.

4. Vout vs. Load Current (10 A $\rightarrow$ 100 A)

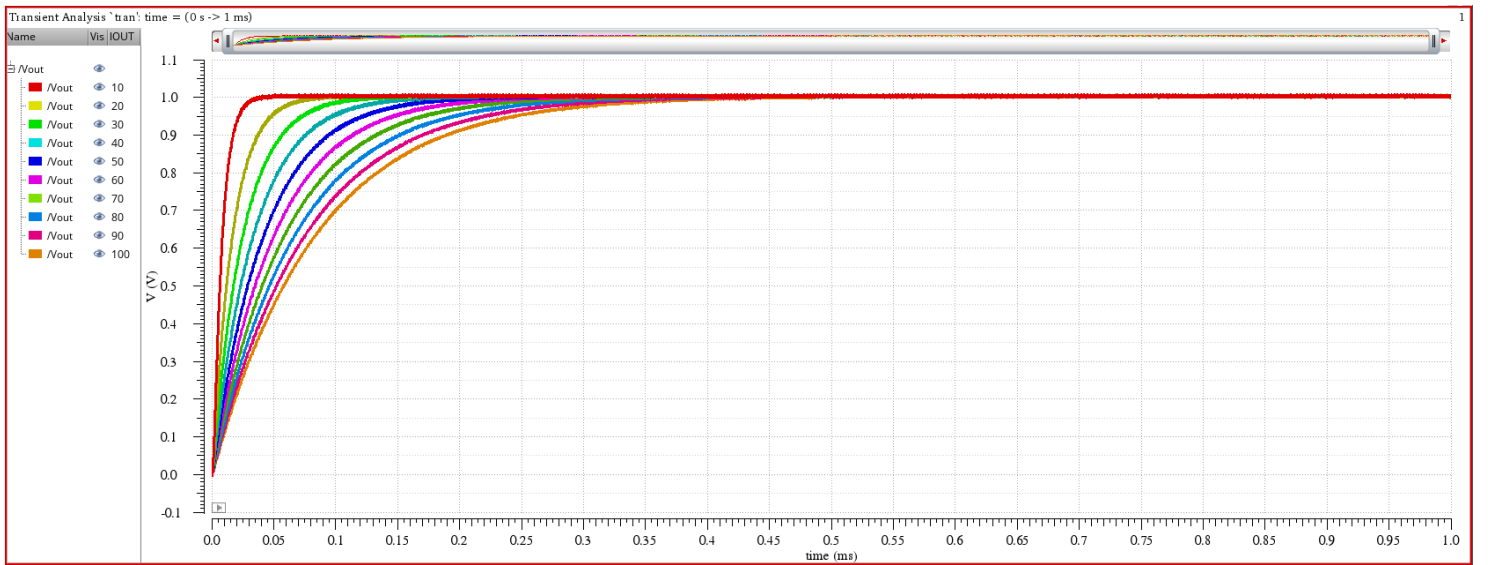


Figure 6. Appended Vout start-up waveforms for a parametric sweep of I_{out} from 10 A to 100 A. X-axis: time (s). Y-axis: Vout (V). Condition: ideal switches, RL swept from 0.1 Ω to 0.01 Ω .

Settling time vs. load current — analysis using the transfer function:

The duty-to-output transfer function of the buck output filter (ideal switches, no ESR) is:

$$G_{vd}(s) = \frac{V_{in}}{1 + s \cdot \frac{L}{RL} + s^2 \cdot LC}$$

which is a standard second-order response with:

$$\omega_n = \frac{1}{\sqrt{LC}} \quad Q = RL \cdot \sqrt{\frac{C}{L}} \quad (\zeta = \frac{1}{2Q})$$

ω_n is fixed by L and C only, so it does not change with load. However, RL sets the quality factor Q, and therefore the damping $\zeta = 1/(2Q)$:

- Light load ($I_{out} = 10\text{ A} \rightarrow RL = 0.1\ \Omega$, larger RL): Q is larger, damping ζ is smaller $\rightarrow$ the response is more underdamped, showing more overshoot/ringing and a longer settling time.
- Heavy load ($I_{out} = 100\text{ A} \rightarrow RL = 0.01\ \Omega$, smaller RL): Q is smaller, damping ζ is larger $\rightarrow$ the load resistor damps the LC resonance more strongly, giving a faster, less oscillatory settling.

This matches the appended waveforms in Figure 6: as I_{out} increases (RL decreases), V_{out} settles faster and with less overshoot, because the load itself provides more damping to the LC filter.

5. IL vs. Load Current (10 A $\rightarrow$ 100 A)

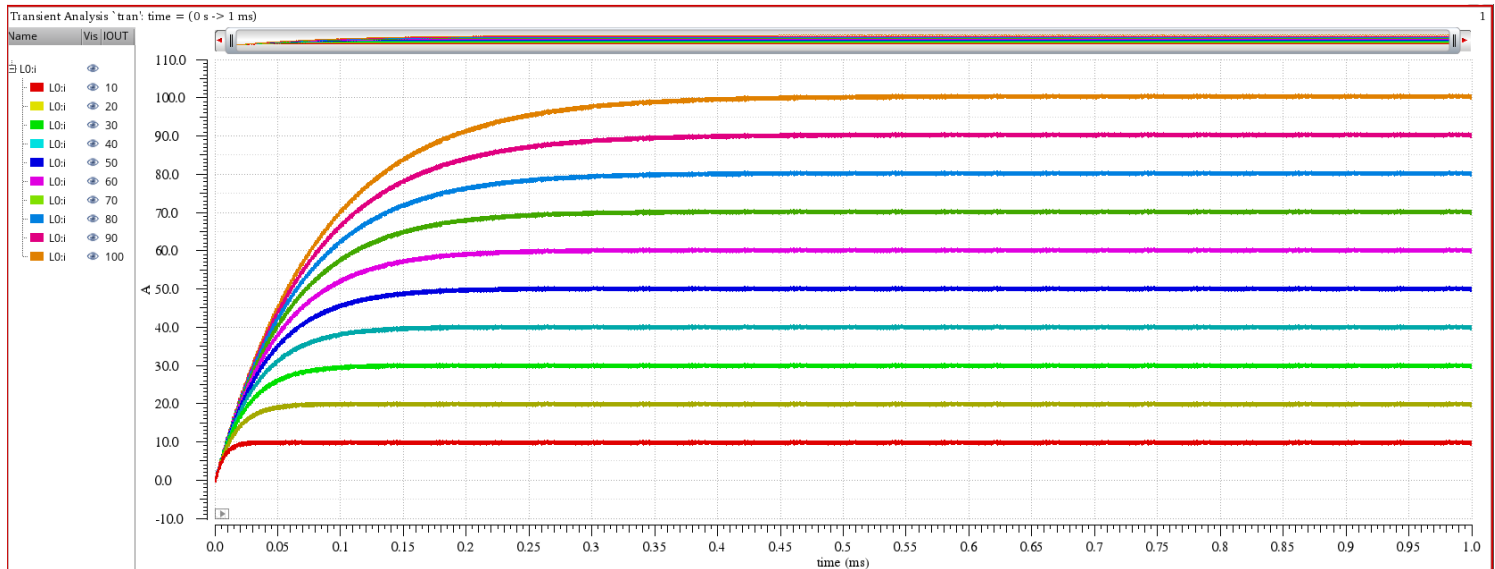


Figure 7. Appended IL start-up waveforms for I_{out} swept from 10 A to 100 A. X-axis: time (s). Y-axis: IL (A). Condition: ideal switches.

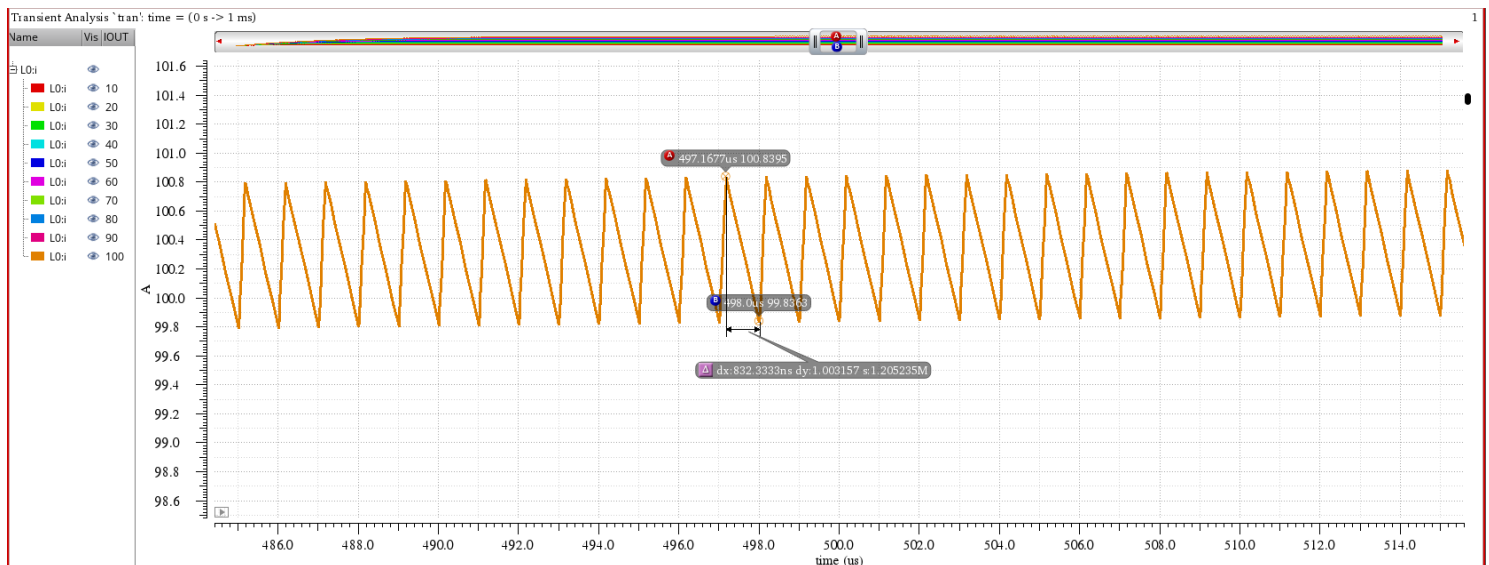


Figure 8. Zoomed steady-state IL ripple. X-axis: time (s). Y-axis: IL (A). Condition: $I_{out} = 100\text{ A}$.

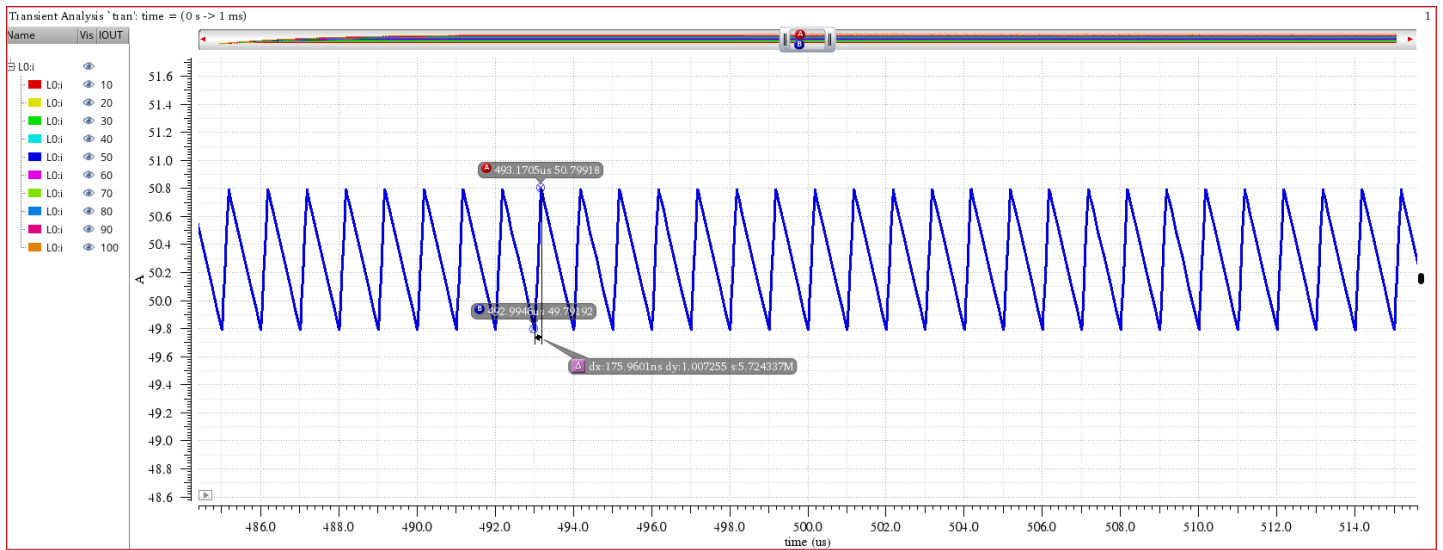


Figure 9. Zoomed steady-state IL ripple. X-axis: time (s). Y-axis: IL (A). Condition: Iout = 50 A.

Comment: The peak-to-peak ripple is almost identical at 100 A and 50 A. With ideal switches ($R_{on} \approx 0$) and no inductor DCR, $\Delta I_{L,pk-pk} = (V_{in} - V_{out}) \cdot D / (L \cdot F_{sw})$ has no dependence on Iout — ripple is set only by Vin, Vout, L and Fsw, not by the load current.

6. Efficiency (Ideal Case)

	IOUT	Pout	Pin	eff
1	10.00	10.13	10.12	100.0
2	20.00	20.25	20.24	100.1
3	30.00	30.38	30.36	100.1
4	40.00	40.50	40.48	100.0
5	50.00	50.62	50.60	100.0
6	60.00	60.73	60.72	100.0
7	70.00	70.84	70.83	100.0
8	80.00	80.92	80.94	99.98
9	90.00	90.96	91.02	99.94
10	100.0	100.9	101.1	99.86

Figure 10. Simulated Pout and Pin, and calculated efficiency at Iout = 100 A.

Assi...	Pout	expr	average(clip(getData("R0:pwr" ?result "tran") 0.0005 0.0006))
Assi...	Pin	expr	average(clip((- getData("V0:pwr" ?result "tran")) 0.0005 0.0006))
Assi...	eff	expr	((Pout / Pin) * 100)

Figure 11. ADE expressions used to compute Pin, Pout and $\eta = P_{out}/P_{in}$.

With ideal switches ($R_{on} \approx 1 \mu\Omega$, $R_{off} \approx 1 T\Omega$) and a lossless L and C, the only dissipation in the circuit is negligible; efficiency comes out essentially 100%, since there is no real resistive element left to dissipate power. This is the theoretical ceiling for a switching converter and is not achievable in practice.

7. Comparison with an Equivalent LDO

An LDO regulates V_{out} by dropping the excess input voltage across a linear pass device, so its theoretical best-case efficiency is simply the conversion ratio:

$$\eta_{LDO} = \frac{V_{out}}{V_{in}} = \frac{1V}{6V} \approx 16.7\%$$

Even in the ideal, lossless limit, an LDO converting 6 V to 1 V can never exceed ~16.7% efficiency, because the remaining 5 V must be dissipated as heat in the pass transistor. The ideal buck converter, by contrast, approaches 100% efficiency because energy is transferred through a lossless LC filter rather than dropped resistively — a ~6× efficiency advantage for this conversion ratio, and the fundamental reason switching regulators are used for large step-down ratios and high current.

Part 2 — Non-Ideal Switches and Inductor DCR

R_{on} of both switches changed to 1 m Ω ; inductor series resistance (DCR) set to 0.5 m Ω .

1. V_{out} at $I_{out} = 100$ A

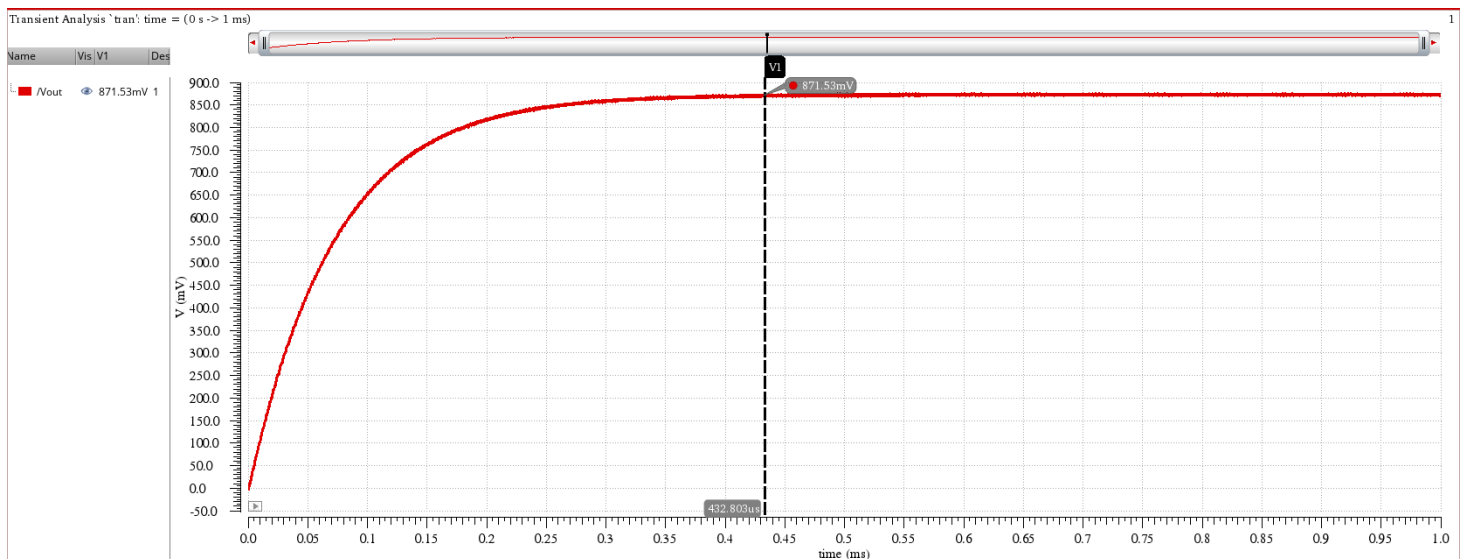


Figure 12. V_{out} transient response with non-ideal switches. X-axis: time (s). Y-axis: V_{out} (V). Condition: $I_{out} = 100$ A, $R_{on} = 1$ m Ω , L DCR = 0.5 m Ω .

Comment: V_{out} no longer settles to 1 V; it settles to a lower value.

Why it does not settle to 1 V:

The high-side switch, low-side switch and inductor DCR now form a real resistive path in series with the load. At 100 A, this path drops:

$$\Delta V \approx I_{out} \times (R_{on_avg} + DCR) \text{ — an IR (conduction-loss) drop that subtracts directly from } V_{out}$$

Since the duty cycle is still set open-loop for the ideal case ($D = 1/6$), the extra resistive drop is not compensated, so V_{out} settles below 1 V. This is a classic load-regulation effect: the higher the load current, the larger the resistive drop and the further V_{out} gets from its target.

2. Vout vs. Load Current (10 A → 100 A)

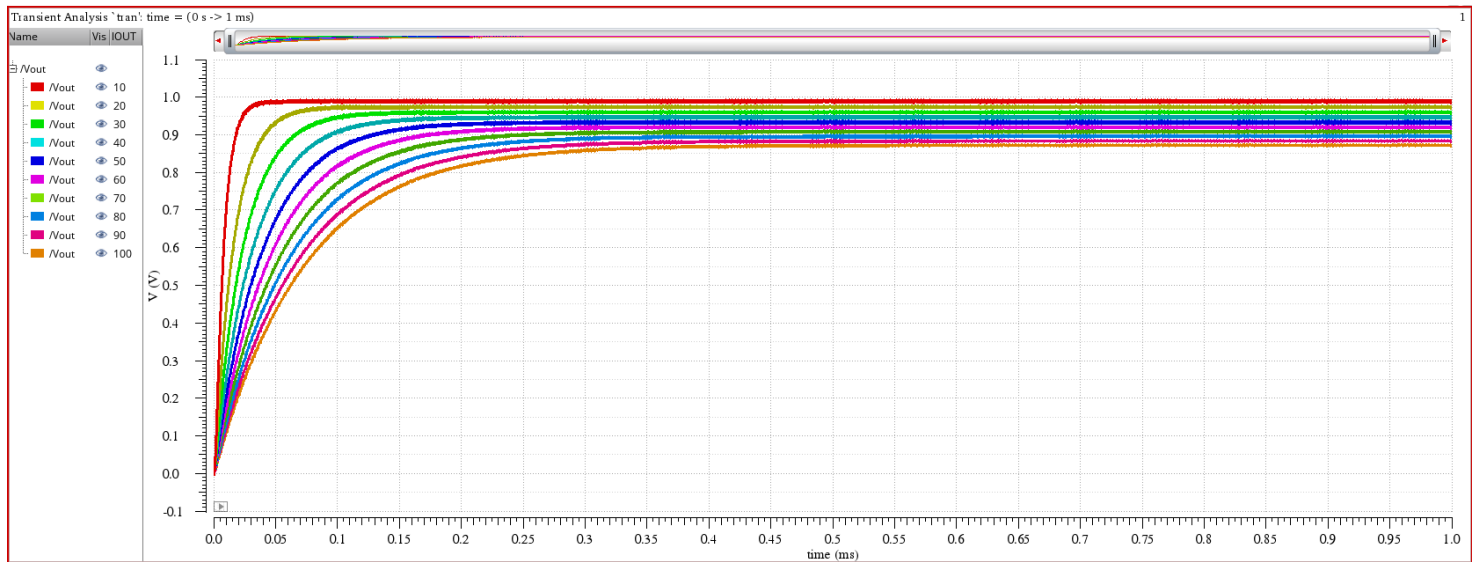


Figure 13. Appended V_{out} waveforms for I_{out} swept from 10 A to 100 A. X-axis: time (s). Y-axis: V_{out} (V). Condition: $R_{on} = 1\text{ m}\Omega$, $L_{DCR} = 0.5\text{ m}\Omega$.

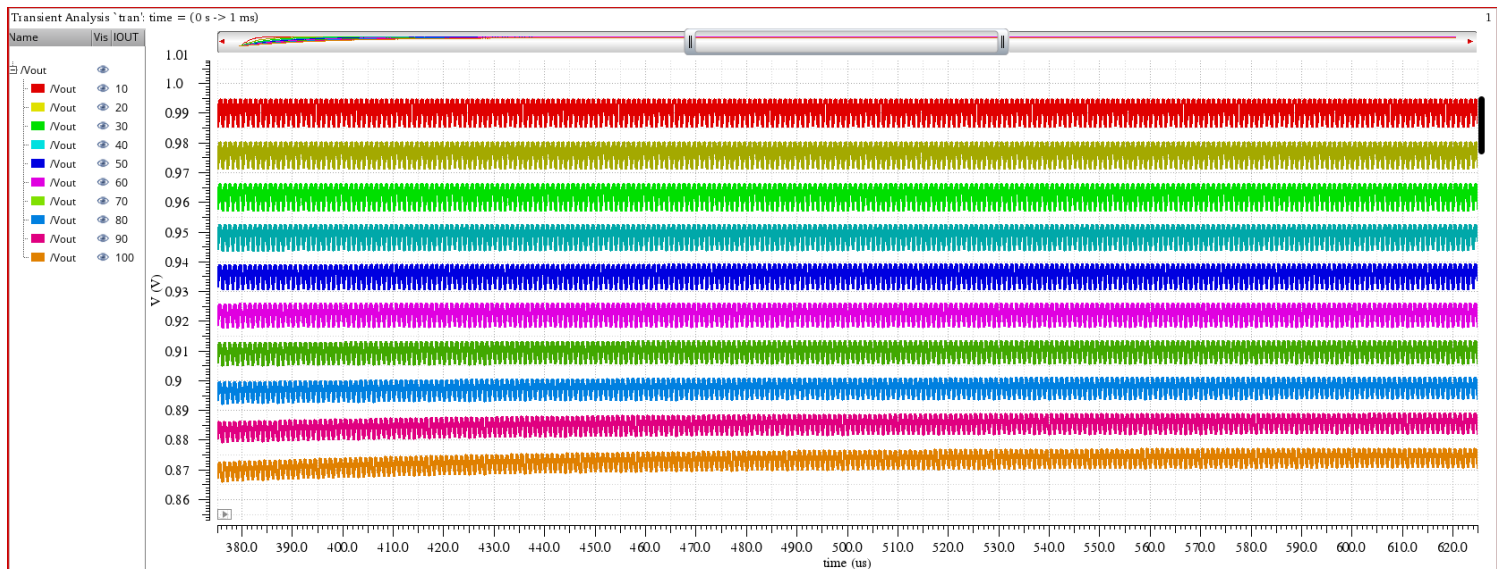


Figure 14. Zoomed steady-state view of the same sweep, showing the spread in final V_{out} across load currents. X-axis: time (s). Y-axis: V_{out} (V).

The voltage drop is smaller at light load and larger at heavy load: at $I_{out} = 10\text{ A}$ the IR drop across R_{on} and D_{CR} is small (V_{out} close to 1 V), while at $I_{out} = 100\text{ A}$ the same resistive path drops ten times more voltage. So the voltage drop decreases in the light-load case, not the heavy-load case.

3. Efficiency vs. Iout

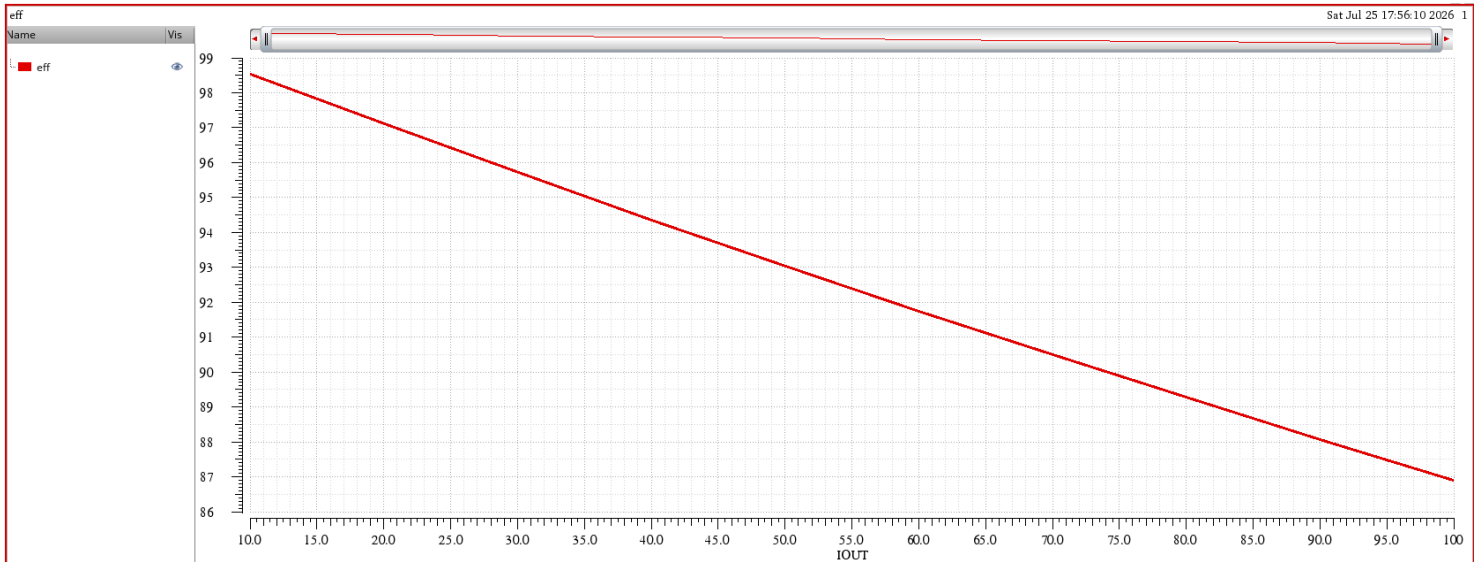


Figure 15. Efficiency vs. output current. X-axis: I_{out} (A), swept 10 A → 100 A. Y-axis: Efficiency (%). Condition: R_{on} = 1 mΩ switches, L DCR = 0.5 mΩ.

Comment: Efficiency drops as I_{out} increases, since conduction losses (I²·R_{on} and I²·DCR) grow with the square of the load current while output power grows only linearly — so losses consume a larger fraction of P_{in} at high current.

4. Replacing the Ideal Switch with a Real Device

A real device was selected to match R_{on} as closely as possible to the target value, and the buck converter was re-designed at a lower voltage/current operating point (V_{in} = 1 V, V_{out} = 0.5 V, I_{out} up to 1 A) so the switch operates within its rated V_{gs}/V_{ds} range.

4.1 Redesign for the Real-Device Operating Point

$$D = \frac{V_{out}}{V_{in}} = \frac{0.5}{1.0} = 0.5$$

$$R_L = \frac{V_{out}}{I_{out,max}} = \frac{0.5}{1} = 0.5 \Omega$$

$$\Delta V_{pk-pk} = 1\% \times 0.5 V = 5 mV$$

$$\Delta I_{pk-pk} = 1\% \times 1 A = 10 mA$$

$$L = (V_{in} - V_{out}) \cdot \frac{D}{\Delta I_{pk-pk} \cdot F_{sw}} = (0.5)(0.5)/(0.01 \times 10^6) = 25 \mu H$$

$$C = \Delta I_{pk-pk} - \frac{pk}{8 \cdot \Delta V_{pk-pk} \cdot F_{sw}} = 0.01/(8 \times 0.005 \times 10^6) = 0.25 \mu F$$

Parameter	Formula	1% ripple value
RL	$V_{out}/I_{out,max}$	$0.5\ \Omega$
$\Delta I_L, pk - pk$	$1\% \times I_{out,max}$	$10\ mA$
$\Delta V_{pk - pk}$	$1\% \times V_{out}$	$5\ mV$
L	$(V_{in} - V_{out}) \cdot D / (\Delta I_{pk - pk} \cdot F_{sw})$	$25\ \mu H$
C	$\Delta I_{pk - pk} / (8 \cdot \Delta V_{pk - pk} \cdot F_{sw})$	$0.25\ \mu F$

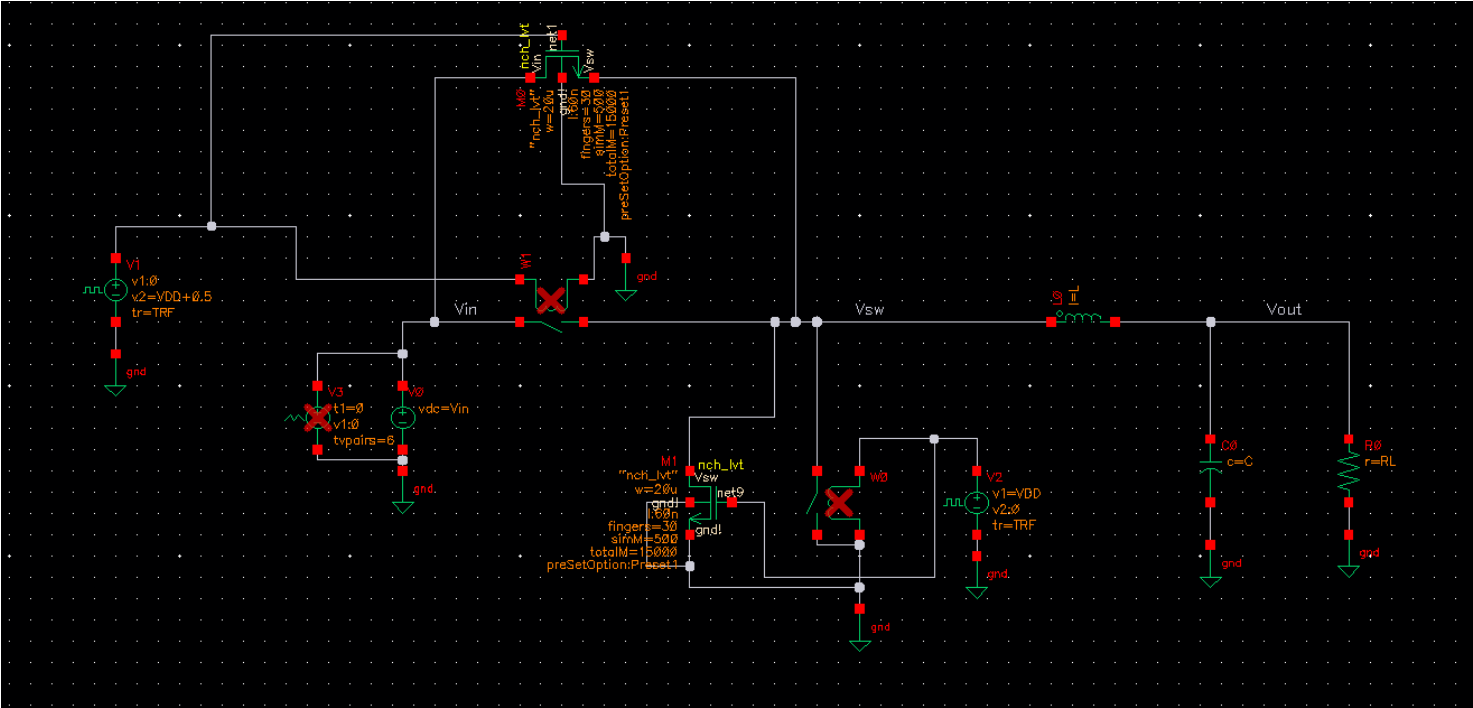


Figure 16. Redesigned buck converter schematic for the real-device operating point ($V_{in} = 1\ V$, $V_{out} = 0.5\ V$, I_{out} up to $1\ A$).

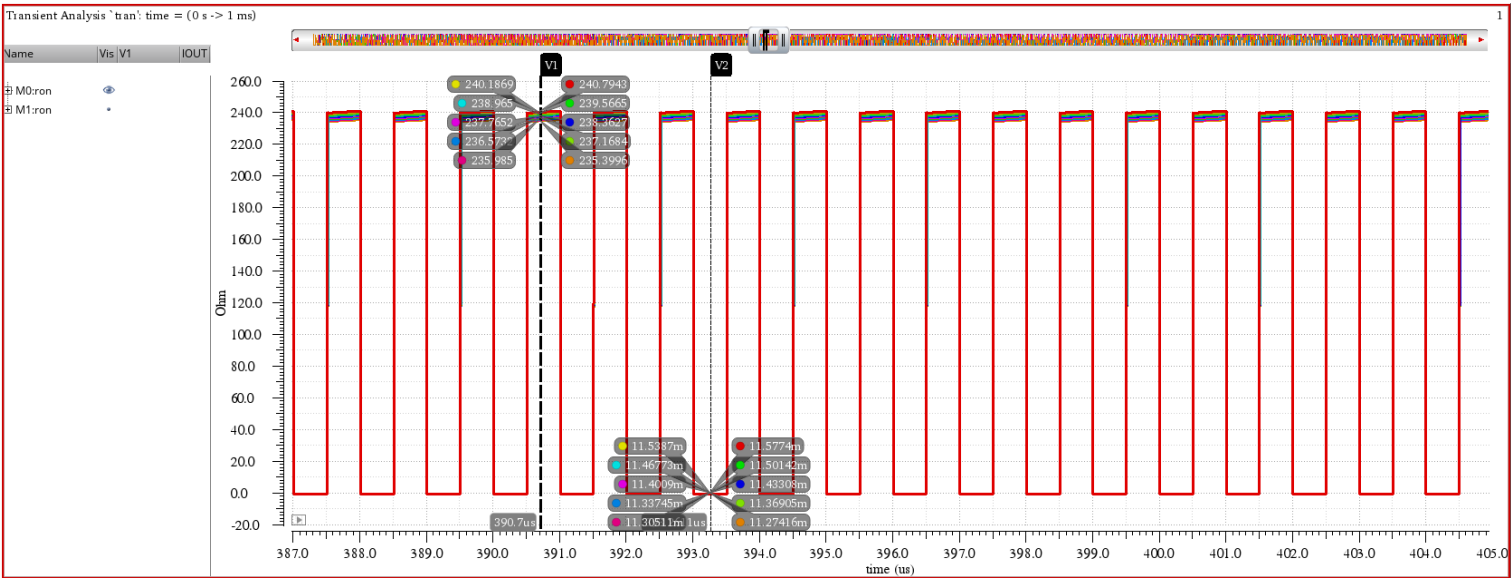


Figure 17. R_{on} extraction / DC sweep used to match the real device's on-resistance to the ideal-switch target.

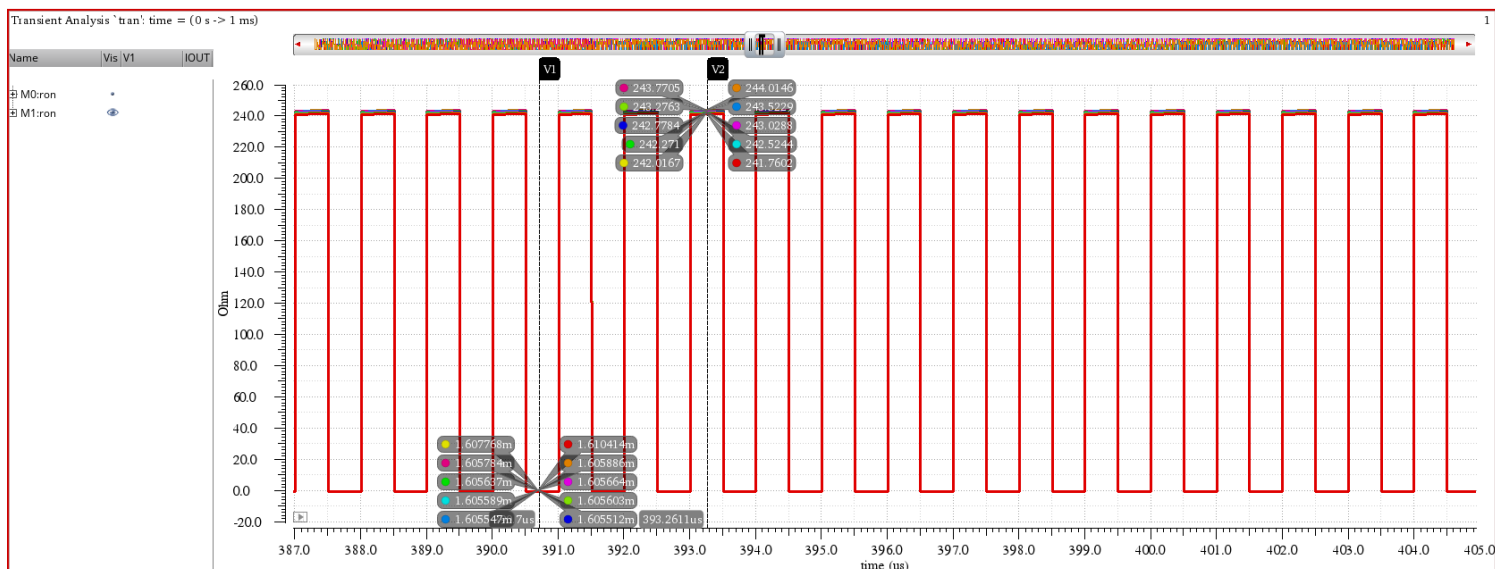


Figure 18. Ron matching for both the high-side and low-side devices.

4.2 Ideal-Switch Reference (at the New Operating Point)

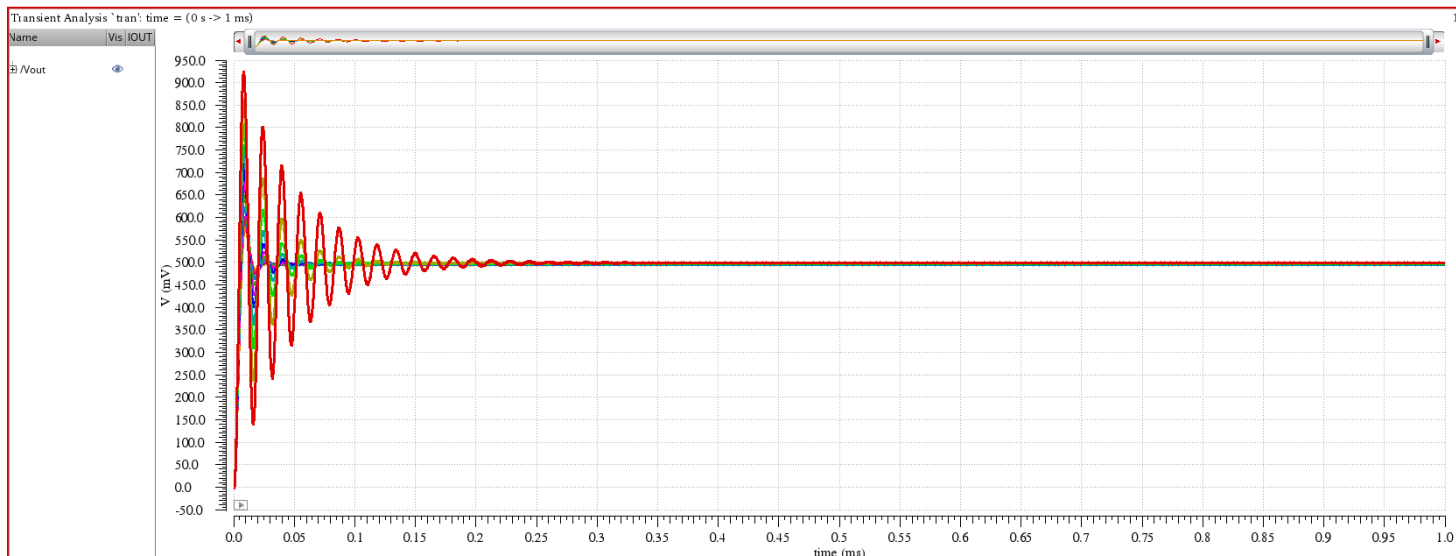


Figure 19. Vout transient response with ideal switches at the new operating point. X-axis: time (s). Y-axis: Vout (V). Condition: Vin = 1 V, Iout up to 1 A.

	IOUT	eff	Pin	Pout
1	100.0E-3	85.51	29.31E-3	25.06E-3
2	200.0E-3	92.09	54.36E-3	50.06E-3
3	300.0E-3	94.48	79.37E-3	74.98E-3
4	400.0E-3	95.68	104.3E-3	99.84E-3
5	500.0E-3	96.40	129.3E-3	124.6E-3
6	600.0E-3	96.85	154.2E-3	149.3E-3
7	700.0E-3	97.17	179.1E-3	174.0E-3
8	800.0E-3	97.39	203.9E-3	198.6E-3
9	900.0E-3	97.54	228.7E-3	223.1E-3
10	1.000	97.65	253.5E-3	247.5E-3

Figure 20. Simulated Pout, Pin and efficiency, ideal switches, new operating point.

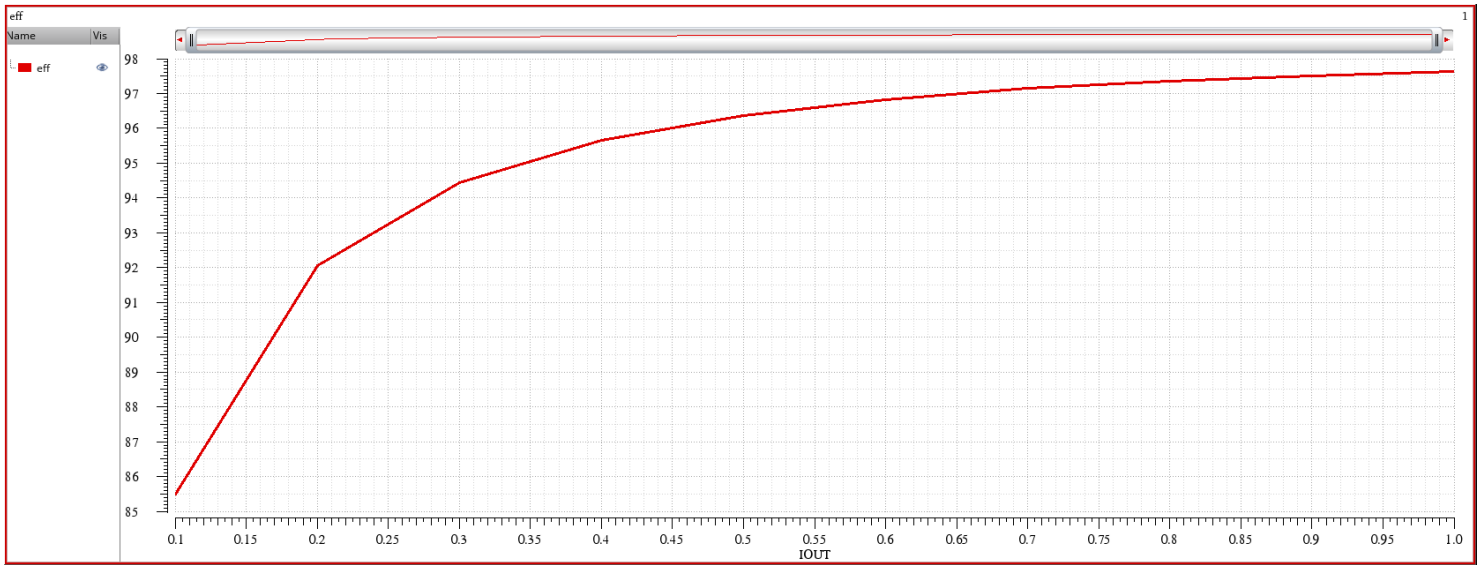


Figure 21. Efficiency vs. Iout, ideal switches. X-axis: Iout (A). Y-axis: Efficiency (%).

Comment: Near-ideal, flat, high efficiency across the current range — the reference baseline for comparison.

4.3 Real-Device Switches

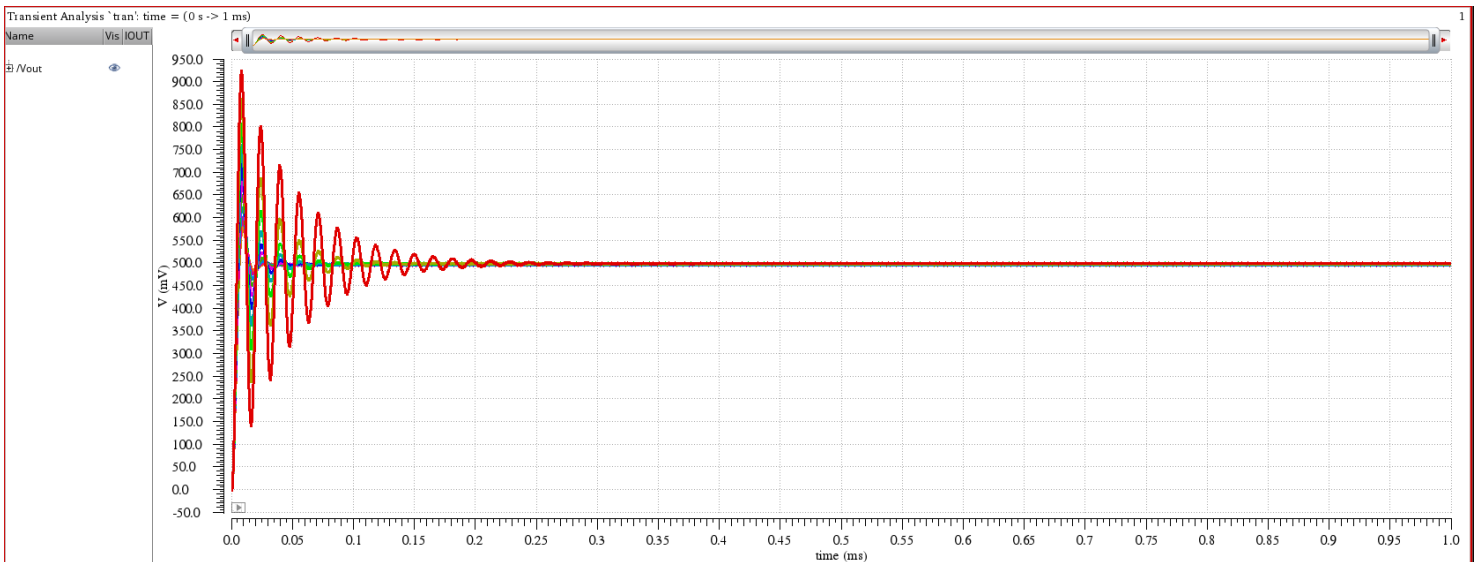


Figure 22. Vout transient response with the real switch device replacing the ideal switches. X-axis: time (s). Y-axis: Vout (V).
Condition: same operating point as 4.2.

	IOUT	eff	Pin	Pout
1	100.0E-3	61.17	40.94E-3	25.04E-3
2	200.0E-3	75.77	66.00E-3	50.01E-3
3	300.0E-3	82.42	90.89E-3	74.91E-3
4	400.0E-3	86.03	115.9E-3	99.75E-3
5	500.0E-3	88.31	141.0E-3	124.5E-3
6	600.0E-3	89.94	165.9E-3	149.2E-3
7	700.0E-3	91.12	190.8E-3	173.8E-3
8	800.0E-3	92.01	215.6E-3	198.4E-3
9	900.0E-3	92.71	240.4E-3	222.9E-3
10	1.000	93.27	265.2E-3	247.3E-3

Figure 23. Simulated Pout, Pin and efficiency with the real device.

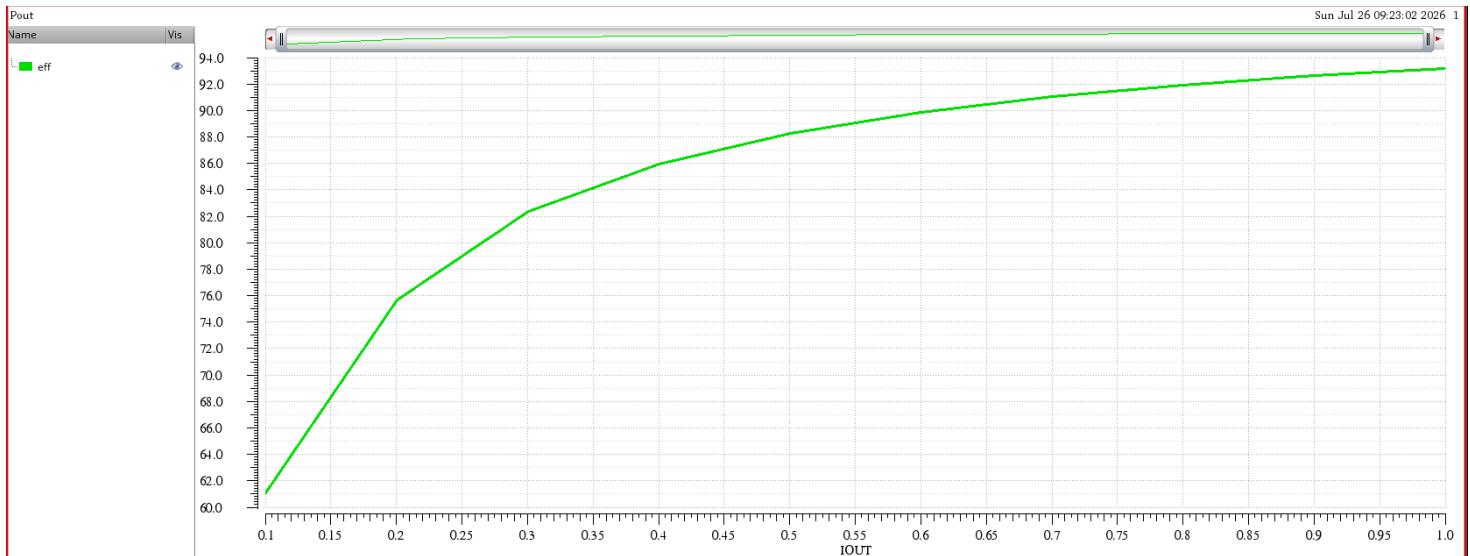


Figure 24. Efficiency vs. I_{OUT} with the real device. X-axis: I_{OUT} (A). Y-axis: Efficiency (%).

Comment: Efficiency is visibly lower than the ideal-switch curve (Figure 21) across the whole current range, and the gap is not constant.

Comparison with the requirement and reasons for the degradation:

Even though the real device's R_{on} was matched as closely as possible to the ideal 1 m Ω target, efficiency still falls short of the Part 2 requirement 3 curve. Reasons:

- R_{on} is not constant in a real device — it varies with V_{gs} , V_{ds} and temperature, so the matched value only holds at one bias point, not across the whole load sweep.
- Switching losses: charging/discharging the device's output capacitance (C_{oss}) and gate capacitance (C_{iss}/Q_g) every switching cycle dissipates power that the ideal switch model does not have.
- Gate-drive loss: driving a real gate charge Q_g at $F_{sw} = 1$ MHz costs $P = Q_g \cdot V_{gs} \cdot F_{sw}$, with no equivalent in the ideal pulse-driven switch.
- Body-diode conduction / reverse recovery during any dead time, which the ideal R_{on}/R_{off} switch model does not exhibit.
- Parasitic package and layout resistance/inductance add further conduction and ringing losses not captured by the ideal model.

Together these explain why the real-device efficiency curve sits below the ideal one, and why simply matching R_{on} is not sufficient to reproduce ideal-switch performance.

Part 3 — Closed-Loop Regulation

Ideal switches are used again for this section.

1. Open-Loop Response to a V_{in} Step

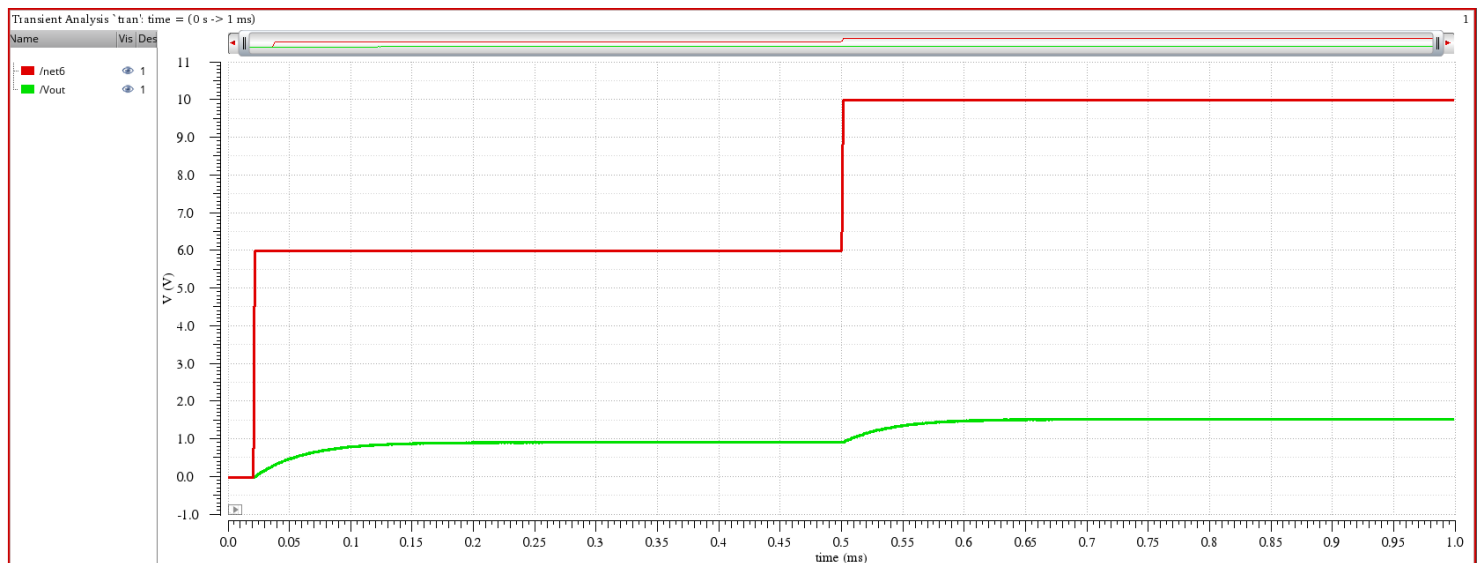


Figure 25. Open-loop response to V_{in} stepped $0\text{ V} \rightarrow 6\text{ V} \rightarrow 10\text{ V}$ within one simulation. X-axis: time (s). Y-axis: V_{in} and V_{out} (V), overlaid. Condition: $I_{out} = 50\text{ A}$, ideal switches, fixed duty cycle (no feedback).

V_{out} tracks V_{in} almost directly (scaled by the fixed duty cycle D), since there is no feedback to correct for the input change. This is not acceptable in real systems: the load expects a regulated 1 V regardless of input-supply variation, and an uncontrolled V_{out} that jumps whenever V_{in} changes could over-voltage or under-voltage downstream circuitry. This motivates closing the loop with feedback.

2. Closed-Loop Response to the Same V_{in} Step

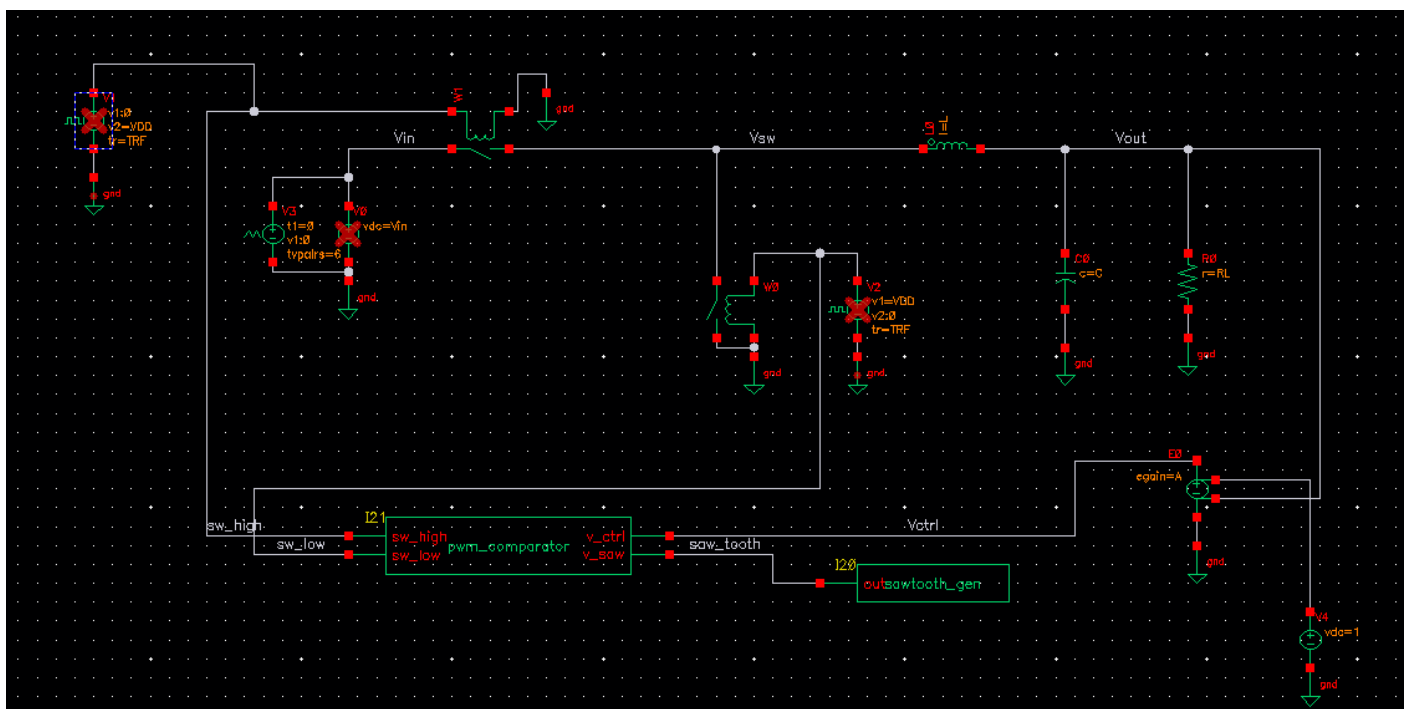


Figure 26. Closed-loop schematic: V_{out} is sensed and compared to V_{ref} through a proportional-gain error amplifier (VCVS), whose output is compared against a sawtooth carrier (PWM comparator, Verilog-A) to generate complementary switch drive signals.

```

output sw_high, sw_low;
electrical v_ctrl, v_saw, sw_high, sw_low;

parameter real vdd = 1.0;          // Switch ON voltage
parameter real vss = 0.0;          // Switch OFF voltage
parameter real trise = 1.0e-9;
parameter real tfall = 1.0e-9;

// Add frequency parameter to calculate max time step locally
parameter real fsw = 1.0e6;

real high_state, low_state;

analog begin
    // Force the simulator to evaluate this math at least 50 times per cycle.
    // This guarantees it will never "skip" the positive part of the comparison.
    $bound_step(1.0 / (fsw * 50.0));

    // Use >= instead of > so that if both are exactly 1V, the switch stays ON
    if (V(v_ctrl) >= V(v_saw)) begin
        high_state = vdd;
        low_state = vss;
    end else begin
        high_state = vss;
        low_state = vdd;
    end

    V(sw_high) <+ transition(high_state, 0, trise, tfall);
    V(sw_low) <+ transition(low_state, 0, trise, tfall);
end
endmodule

```

Figure 27. Verilog-A source for the PWM comparator: compares the error-amplifier output to the sawtooth ramp and outputs the complementary high-side/low-side switch drive signals.

```

// VerilogA for Assignments, sawtooth_gen, veriloga

`include "constants.vams"
`include "disciplines.vams"

module sawtooth_gen (out);
    output out;
    electrical out;

    parameter real freq = 1.0e6;    // 1 MHz
    parameter real vlow = 0.0;
    parameter real vhigh = 1.0;
    parameter real tfall = 1.0e-9;  // 1 ns reset time

    real period, trise, t_rel;

    analog begin
        period = 1.0 / freq;
        trise = period - tfall;

        // Force Spectre to take at least 50 steps per period
        // This guarantees the simulation never gets "stuck"
        $bound_step(period / 50.0);

        // Continuous mathematical modulo
        t_rel = $abstime - floor($abstime / period) * period;

        if (t_rel < trise)
            V(out) <+ vlow + (vhigh - vlow) * (t_rel / trise);
        else
            V(out) <+ vhigh - (vhigh - vlow) * ((t_rel - trise) / tfall);
    end
endmodule

```

Figure 28. Verilog-A source for the sawtooth carrier generator ($0 \rightarrow 1$ V, period = $1/F_{sw} = 1 \mu s$).

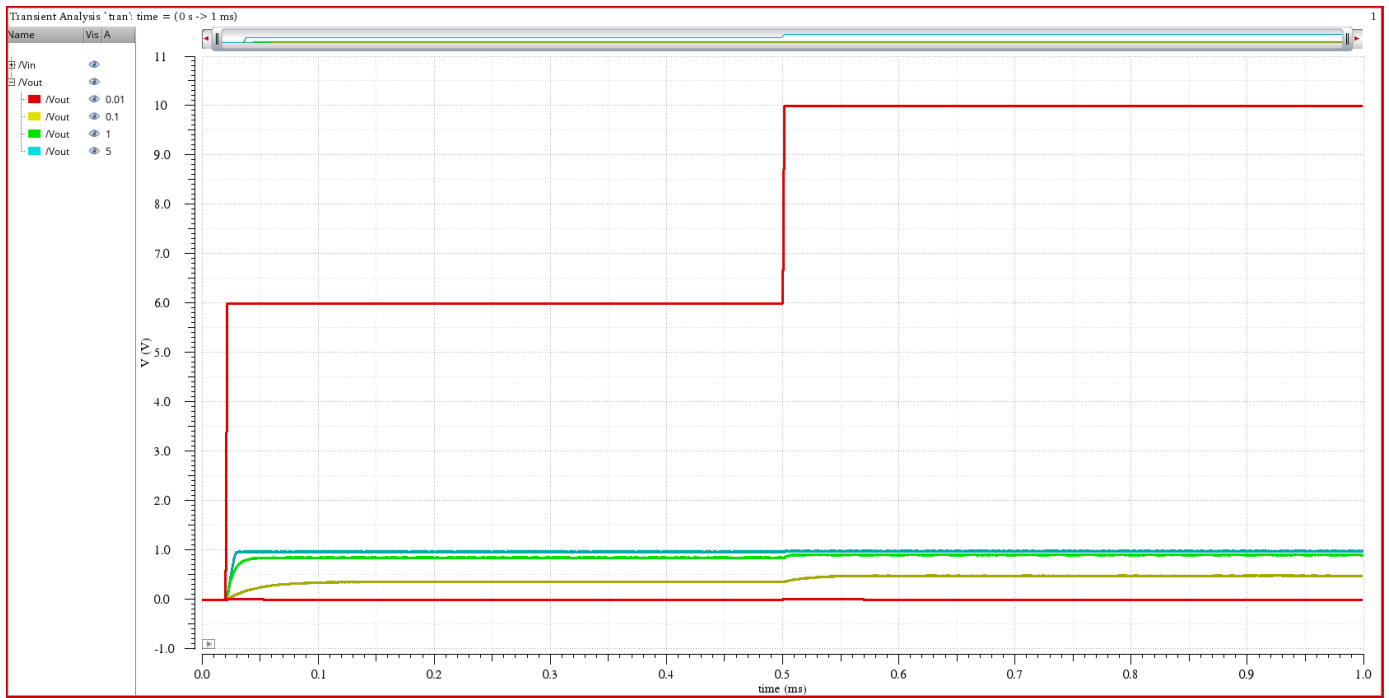


Figure 29. *Vout* waveform at *I*_{out} = 50 A for the same *V*_{in} step (0 → 6 → 10 V), closed loop, for proportional gain *A*_v = 0.01, 0.1, 1 and 5, overlaid with the open-loop case. X-axis: time (s). Y-axis: *V*_{out} (V).

Speed and rejection vs. *A*_v:

- *A*_v = 0.01 and 0.1: the loop barely corrects the output — the response is close to the open-loop curve, with large steady-state error and slow, weak rejection of the *V*_{in} step. Gain is too small to move ctrl meaningfully relative to the sawtooth.
- *A*_v = 1: clearly faster correction and much smaller deviation from 1 V after each *V*_{in} step; line rejection is noticeably improved over the low-gain cases.
- *A*_v = 5: fastest response and smallest steady-state error of the four — the loop rejects the *V*_{in} step almost immediately, since a proportional controller's steady-state error scales as $\sim 1/A_v$. Very high gain does, however, raise the risk of ringing/instability, so there is a practical trade-off, not a free win, as *A*_v keeps increasing.

Overall: increasing *A*_v trades faster settling and better rejection of input variations for higher risk of overshoot/instability, illustrating the classic proportional-control speed vs. steady-state-error vs. stability trade-off, and why closed-loop control is required to keep *V*_{out} regulated as *V*_{in} (or the load) varies.

Appendix

Name	Type	Details
Pout	expr	average(clip(getData("R0:pwr" ?result "tran") 0.0005 0.0006))
Pin	expr	average(clip((- getData("V0:pwr" ?result "tran")) 0.0005 0.0006))
eff	expr	((Pout / Pin) * 100)

Figure 30. ADE expressions used for the efficiency and other post-processed measurements throughout this report.